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Step 5 - CNTFRQ and Time Conversion

Reference Note

This chapter is based on the ARM Generic Timer Architecture.

The System Counter runs at a fixed frequency. Software uses **CNTFRQ** to determine that frequency and convert between:

Time <-> Counter Ticks

Learning Objectives

After this chapter you should understand:

- Purpose of CNTFRQ
- Frequency units
- Time-to-tick conversion
- Tick-to-time conversion
- Practical timer calculations

1. What is CNTFRQ?

`CNTFRQ` contains the System Counter frequency.

Example:

Register	Value
CNTFRQ	24,000,000

Meaning:

System Counter increments
24,000,000 times per second

2. Why Is It Needed?

Suppose:

Counter Value = 48,000,000

Without frequency information:

48,000,000 means nothing

With:

CNTFRQ = 24,000,000

we can determine:

Elapsed Time = 2 seconds

3. Common Frequencies

Frequency	Common Usage
19.2 MHz	Mobile SoCs
24 MHz	Embedded SoCs
25 MHz	Networking Devices
50 MHz	Some server platforms

4. Time to Tick Conversion

Formula:

Ticks = Time × Frequency

Example

Parameter	Value
Frequency	24 MHz
Time	1 ms

Calculation:

Ticks
=
24,000,000 × 0.001
=
24,000

Result:

1 ms = 24,000 ticks

5. Tick to Time Conversion

Formula:

Time = Ticks / Frequency

Example

Parameter	Value
Ticks	48,000
Frequency	24 MHz

Calculation:

48,000 / 24,000,000
=
0.002 seconds

Result:

2 ms

Quick Reference Table

24 MHz Counter

Time	Ticks
1 us	24
10 us	240
100 us	2,400
1 ms	24,000

10 ms	240,000
100 ms	2,400,000
1 second	24,000,000

Key Takeaways

Concept	Description
CNTFRQ	System Counter frequency
Needed For	Time conversion
Formula	Ticks = Time × Frequency
Formula	Time = Ticks ÷ Frequency

Knowledge Check

Q1

What does CNTFRQ contain?

Q2

How many ticks occur in 1 ms at 24 MHz?

Q3

How many milliseconds are represented by 48,000 ticks at 24 MHz?

Answer Key

Question	Answer
Q1	System Counter frequency
Q2	24,000

Q3	2 ms